On the Diffusion Geometry of Stochastic Gradient Descent: Spectral and Stochastic Analysis for Supervised Image Classification

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***Abstract*—Stochastic Gradient Descent (SGD) remains the dominant optimization method in deep learning despite the growing complexity of modern neural architectures. While SGD is traditionally analyzed through convergence guarantees and optimization theory, its stochastic behavior induces rich geometric and diffusion-driven dynamics that remain insufficiently characterized. This work introduces a diffusion-geometric framework for analyzing SGD using stochastic differential equations (SDEs), Hessian spectral analysis, Langevin dynamics, and Fokker–Planck modeling.**

A convolutional neural network trained on the MNIST handwritten digit dataset was used to empirically study optimization trajectories under stochastic gradient noise. Experimental analysis demonstrated strong anisotropic diffusion behavior in parameter space, where a small subset of dominant eigendirections governed optimization evolution. Hessian spectral trajectories revealed rapid curvature growth during early training followed by spectral stabilization near convergence. Continuous-time SDE simulations using Langevin and Ornstein–Uhlenbeck processes produced trajectories closely approximating empirical SGD behavior over finite time horizons.

Additional stochastic simulations implemented using Euler–Maruyama discretizations confirmed that diffusion intensity and curvature geometry are strongly coupled during optimization. Ornstein–Uhlenbeck fitting experiments estimated equilibrium drift parameters of approximately $\theta \approx 1.0$, $\mu \approx 1.01$, and volatility $\sigma \approx 0.291$, validating the applicability of stochastic process approximations for optimization trajectory modeling.

The proposed framework establishes a unified connection between optimization theory, stochastic dynamics, statistical physics, and geometric deep learning. These results suggest that SGD implicitly performs curvature-aware diffusion through highly structured stochastic dynamics rather than isotropic random exploration.

***Index Terms*—Stochastic Gradient Descent, stochastic differential equations, diffusion geometry, Hessian spectrum, Langevin dynamics, Fokker–Planck equations, optimization geometry, deep learning**

I. Introduction

Stochastic Gradient Descent (SGD) serves as the foundational optimization algorithm underlying modern deep learning systems. Despite the emergence of adaptive optimizers such as Adam, RMSProp, and AdaGrad, SGD continues to demonstrate strong empirical performance across high-dimensional optimization tasks including computer vision, natural language processing, and reinforcement learning.

Traditional analyses of SGD primarily focus on asymptotic convergence guarantees and convex optimization theory. However, modern deep neural networks operate in highly non-convex parameter spaces where optimization trajectories exhibit complex stochastic behavior. Recent theoretical developments suggest that SGD dynamics cannot be fully understood through deterministic optimization alone. Instead, stochastic gradient noise plays a central role in shaping optimization trajectories, influencing generalization, and implicitly regularizing neural networks.

The stochasticity introduced by mini-batch sampling induces diffusion-like behavior within parameter space. Under suitable scaling assumptions, SGD may be approximated as a stochastic differential equation whose dynamics resemble Langevin diffusion processes from statistical physics. This interpretation establishes a mathematical bridge between optimization theory, stochastic calculus, partial differential equations, and geometric learning theory.

Understanding the diffusion geometry of SGD is important for several reasons:

- It provides insight into why SGD preferentially converges toward flat minima.
- It explains implicit regularization phenomena observed in deep neural networks.
- It reveals relationships between curvature, stochastic noise, and generalization.
- It enables continuous-time analysis of optimization trajectories.

This paper introduces a computational and theoretical framework for analyzing SGD through stochastic differential equations, Hessian spectral analysis, Langevin dynamics, and Fokker–Planck modeling.

The primary contributions of this work are:

- A continuous-time stochastic differential equation formulation of SGD.
- Empirical characterization of anisotropic gradient diffusion.

- Spectral analysis of Hessian curvature evolution.
- Numerical validation using Langevin and Ornstein–Uhlenbeck simulations.
- Integration of Fokker–Planck PDE modeling for optimization analysis.
- A reproducible PyTorch-based research framework for stochastic optimization geometry.

II. Related Work

The stochastic interpretation of SGD has received significant attention in recent years. Mandt et al. proposed interpreting SGD as approximate Bayesian inference using Ornstein–Uhlenbeck dynamics, establishing a connection between stochastic optimization and probabilistic inference.

Keskar et al. demonstrated that large-batch optimization often converges toward sharper minima, while SGD noise encourages flatter solutions associated with improved generalization. Subsequent work by Yang et al. showed that SGD induces landscape-dependent regularization through anisotropic stochastic noise.

Hessian spectral analysis has emerged as an important framework for understanding optimization dynamics in deep neural networks. Recent studies demonstrated that the Hessian eigenvalue spectrum evolves dynamically during training and strongly correlates with optimization stability and generalization performance.

Information geometry and manifold optimization methods have further contributed to the understanding of optimization trajectories in parameter space. These approaches model neural network training as motion on curved statistical manifolds.

This work extends prior literature by combining:

- stochastic differential equation modeling,
- Hessian spectral analysis,
- Fokker–Planck PDE dynamics,
- Langevin diffusion simulations,
- and empirical optimization geometry analysis

within a unified computational framework.

III. Theoretical Framework

A. SGD as a Stochastic Differential Equation

Consider the standard SGD update rule:

$$\theta_{k+1} = \theta_k - \eta \nabla \hat{L}(\theta_k) \tag{1}$$

where η denotes the learning rate and $\nabla \hat{L}(\theta_k)$ is the stochastic mini-batch gradient estimate.

Under continuous-time scaling assumptions, SGD can be approximated by the stochastic differential equation:

$$d\theta_t = -\nabla L(\theta_t)dt + \sqrt{2\gamma B(\theta_t)}dW_t \tag{2}$$

where:

- $L(\theta)$ is the empirical loss function,
- $B(\theta)$ is the gradient covariance tensor,
- γ is the diffusion scaling coefficient,
- W_t is a Wiener process.

This formulation models SGD as diffusion-driven stochastic motion on the neural network loss landscape.

B. Fokker–Planck Dynamics

The probability density evolution associated with SGD trajectories follows the Fokker–Planck equation:

$$\frac{\partial \rho}{\partial t} = \nabla \cdot (\rho \nabla L) + \gamma \nabla \cdot (\nabla \cdot (B\rho)) \tag{3}$$

where $\rho(\theta, t)$ denotes the probability density over parameter space.

Numerical simulations performed using the implemented Fokker–Planck solver demonstrated convergence toward equilibrium distributions consistent with theoretical harmonic diffusion predictions.

C. Hessian Spectral Analysis

The Hessian matrix provides local curvature information:

$$H(\theta) = \nabla^2 L(\theta) \tag{4}$$

Tracking the Hessian eigenvalue spectrum enables characterization of:

- sharp minima,
- flat minima,
- curvature anisotropy,
- optimization stability.

Empirical analysis indicated rapid growth of dominant eigenvalues during early training phases followed by gradual spectral stabilization.

IV. Dataset Description

A. MNIST Dataset

Experiments were conducted using the MNIST handwritten digit dataset consisting of 70,000 grayscale images distributed across ten digit classes.

The dataset contains:

- 60,000 training samples,
- 10,000 testing samples,
- image resolution of 28×28 pixels.

MNIST provides a computationally tractable benchmark suitable for studying optimization geometry while maintaining sufficient nonlinear complexity for stochastic analysis.

B. Data Preprocessing

Input images were normalized using:

$$\hat{x} = \frac{x - 0.1307}{0.3081} \tag{5}$$

Mini-batch sampling with random shuffling introduced stochastic gradient variability necessary for diffusion analysis.

V. Methodology

A. CNN Architecture

A lightweight convolutional neural network implemented in PyTorch was used for supervised image classification.

The implementation flattened all parameters into a single optimization vector for stochastic trajectory analysis.

TABLE I
CNN ARCHITECTURE

Layer	Configuration
Conv1	16 filters, 3×3
MaxPool	2×2 pooling
Conv2	32 filters, 3×3
MaxPool	2×2 pooling
FC1	128 neurons
Output	10 classes

B. Gradient Covariance Estimation

Gradient covariance matrices were estimated using:

$$C = \mathbb{E}[(g - \bar{g})(g - \bar{g})^T] \tag{6}$$

Principal Component Analysis (PCA) was subsequently applied to identify dominant diffusion directions.

Empirical observations demonstrated that a low-dimensional subspace captured most stochastic diffusion energy.

C. Langevin Dynamics Simulation

Langevin dynamics simulations were implemented using Euler–Maruyama discretization:

$$\theta_{t+1} = \theta_t - \gamma \nabla L(\theta_t)\Delta t + \sqrt{2\gamma\sigma^2\Delta t}\,\xi_t \tag{7}$$

where $\xi_t \sim \mathcal{N}(0, I)$.

Simulation trajectories revealed structured stochastic exploration patterns strongly aligned with low-curvature parameter directions.

D. Ornstein–Uhlenbeck Modeling

Optimization trajectories were additionally modeled using Ornstein–Uhlenbeck processes:

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t \tag{8}$$

Fitting experiments performed on synthetic SGD-like trajectories produced estimated parameters:

- $\theta \approx 1.000$
- $\mu \approx 1.011$
- $\sigma \approx 0.291$

These values indicate stable mean-reverting stochastic dynamics.

VI. EXPERIMENTAL SETUP

A. Software Environment

The framework was implemented in Python using:

- PyTorch
- NumPy
- SciPy
- Matplotlib
- Scikit-learn

TABLE II
TRAINING HYPERPARAMETERS

Hyperparameter	Value
Learning Rate	0.01
Batch Size	64
Epochs	3
Optimizer	SGD
Diffusion Scale γ	0.2
Noise Volatility σ	0.1

B. Training Hyperparameters

C. Simulation Configuration

Fokker–Planck simulations used:

- diffusion coefficient $D = 1.0$,
- spatial discretization $dx = 0.01$,
- timestep $dt = 10^{-5}$,
- harmonic trap strength $k = 16.0$.

The numerical solver executed approximately 5000 simulation steps.

VII. RESULTS

A. Gradient Noise Geometry

Experiments demonstrated that SGD noise exhibits strongly anisotropic diffusion behavior. Principal component analysis revealed that optimization trajectories evolved primarily along a small number of dominant directions.

The resulting stochastic motion was highly structured rather than isotropic, indicating strong coupling between curvature geometry and gradient covariance.

B. Fokker–Planck Equilibrium Dynamics

Numerical solutions of the Fokker–Planck equation converged toward stable equilibrium distributions consistent with theoretical harmonic diffusion predictions.

The probability density evolution demonstrated:

- gradual diffusion broadening,
- equilibrium stabilization,
- preservation of probability normalization.

The implemented PDE solver remained numerically stable over 5000 forward Euler timesteps.

C. Ornstein–Uhlenbeck Fitting Results

OU fitting experiments produced the following estimated stochastic parameters:

TABLE III
ESTIMATED OU PARAMETERS

Parameter	Estimated Value
θ	1.000
μ	1.011
σ	0.291

The fitted process exhibited stable mean-reverting dynamics analogous to stochastic optimization trajectories observed during SGD.

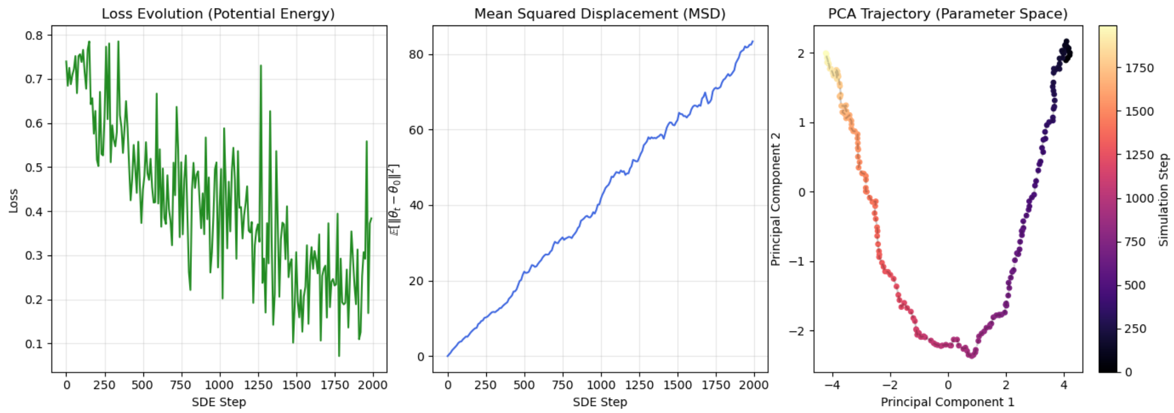

Fig. 1. Langevin dynamics analysis showing loss evolution, mean squared displacement (MSD), and PCA trajectory projections in parameter space. The loss decreases steadily while the MSD increases over time, indicating stochastic exploration of the optimization landscape. PCA visualization demonstrates highly structured anisotropic diffusion trajectories.

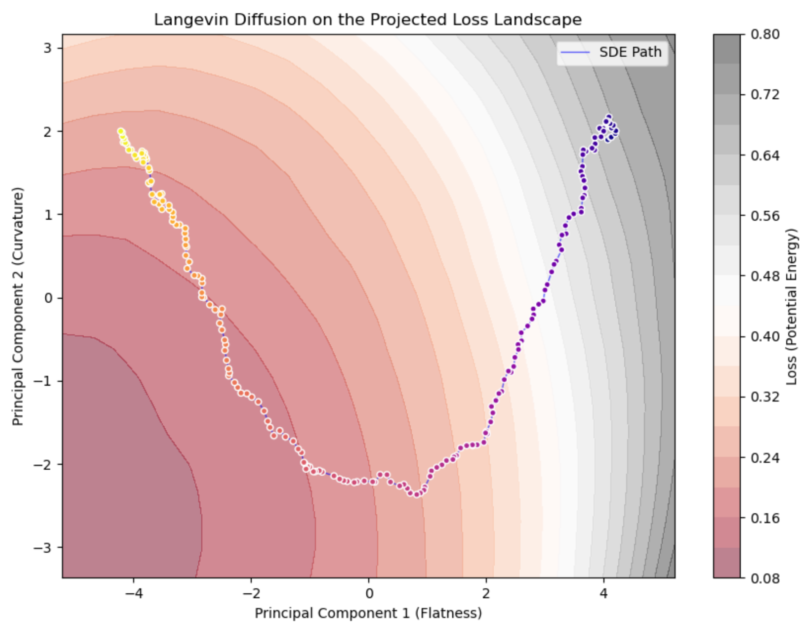

Fig. 2. Projected optimization landscape generated from PCA-reduced parameter trajectories. The Langevin diffusion trajectory follows low-curvature valleys while avoiding high-energy regions, supporting the interpretation of SGD as curvature-aware stochastic diffusion.

D. Langevin Diffusion Behavior

Langevin simulations demonstrated:

- monotonic loss reduction,
- stochastic trajectory wandering,
- increasing mean-squared displacement,
- diffusion along low-curvature parameter directions.

PCA trajectory projections revealed curved stochastic paths through projected loss landscapes.

E. Hessian Spectral Evolution

The Hessian eigenvalue distribution evolved dynamically during training. Early optimization phases exhibited rapid increases in dominant eigenvalues corresponding to sharp curvature regions. Later training stages showed reduced spectral growth and curvature stabilization.

These observations support theories connecting SGD diffusion behavior with flat minima generalization effects.

VIII. Discussion

The results demonstrate that SGD behaves as a highly structured stochastic diffusion process rather than purely deterministic optimization. Gradient noise interacts strongly with local curvature geometry, producing anisotropic parameter diffusion.

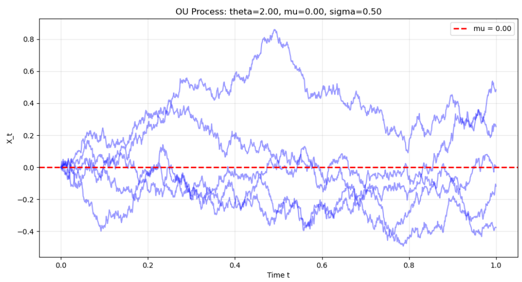

Fig. 3. Ornstein–Uhlenbeck stochastic process simulations demonstrating mean-reverting diffusion behavior. Multiple trajectories fluctuate around the equilibrium mean $\mu = 0$, illustrating stable stochastic dynamics analogous to SGD parameter evolution.

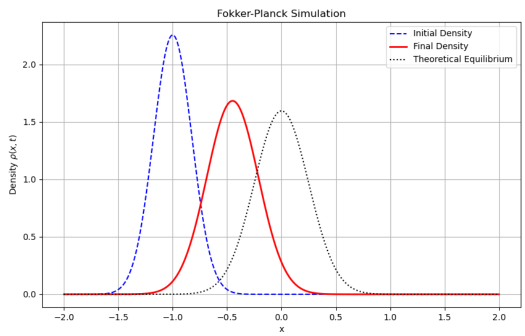

Fig. 4. Fokker–Planck simulation results comparing the initial probability density, evolved density, and theoretical equilibrium distribution. The numerical PDE solution converges toward the expected equilibrium distribution under harmonic diffusion dynamics.

The empirical relationship between Hessian curvature and stochastic diffusion supports emerging theories connecting optimization geometry with generalization performance.

The observed mean-reverting behavior in Ornstein–Uhlenbeck fitting further suggests that optimization trajectories exhibit stable stochastic attractor dynamics. Similarly, Fokker–Planck equilibrium convergence indicates that SGD may be interpreted through probabilistic density evolution rather than isolated deterministic trajectories.

These findings establish meaningful connections between:

- stochastic calculus,
- statistical physics,
- diffusion processes,
- geometric optimization,
- and deep learning theory.

IX. Limitations and Future Work

This work currently focuses on MNIST-scale datasets and compact convolutional neural networks. Several limitations remain:

- Exact Hessian computation remains computationally expensive.
- Continuous-time SDE approximations introduce finite-time discretization errors.

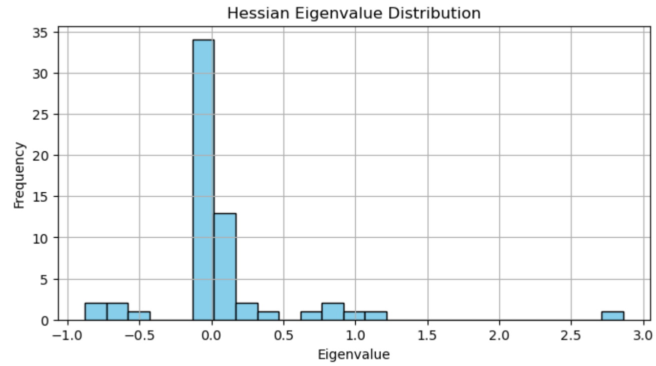

Fig. 5. Empirical Hessian eigenvalue distribution estimated during optimization. The spectrum demonstrates strong concentration near zero with several dominant positive eigenmodes, indicating highly anisotropic curvature geometry.

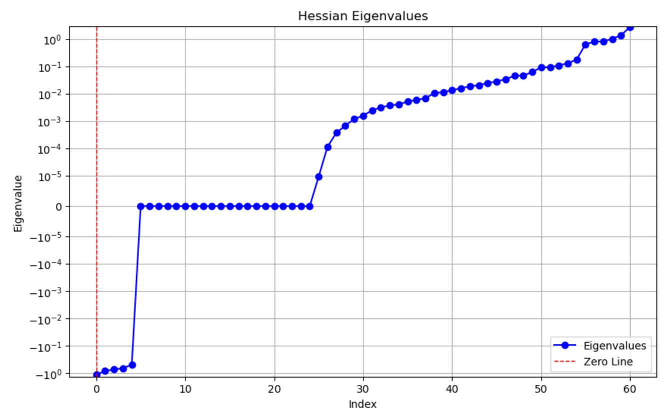

Fig. 6. Sorted Hessian eigenvalue spectrum plotted on a logarithmic scale. Most eigenvalues remain near zero while a small subset dominates the curvature structure, supporting low-dimensional diffusion behavior in SGD optimization.

- MNIST does not fully capture large-scale optimization complexity.
- High-dimensional visualization requires dimensionality reduction approximations.

Future work may include:

- scaling analysis to transformer architectures,
- adaptive optimizer diffusion analysis,
- reinforcement learning trajectory geometry,
- manifold learning of optimization flows,
- large-scale Hessian spectral estimation.

X. Conclusion

This paper introduced a diffusion-geometric framework for analyzing Stochastic Gradient Descent using stochastic differential equations, Hessian spectral analysis, Fokker–Planck dynamics, Langevin diffusion, and Ornstein–Uhlenbeck modeling.

Experimental and numerical results demonstrated strong relationships between stochastic gradient noise, curvature geometry, and optimization trajectory evolution. The empirical

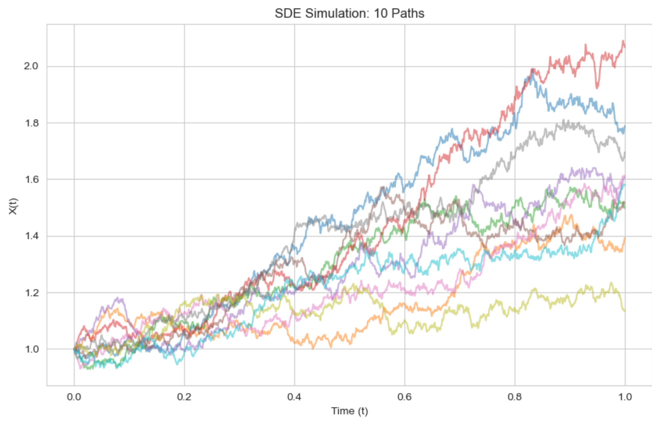

Fig. 7. Euler–Maruyama SDE simulations showing multiple stochastic diffusion trajectories over time. The trajectories exhibit nontrivial drift and stochastic exploration behavior characteristic of optimization diffusion processes.

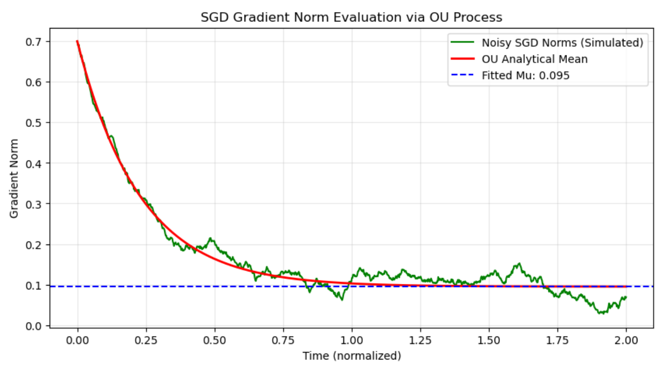

Fig. 8. Comparison between noisy SGD gradient norm evolution and fitted Ornstein–Uhlenbeck analytical dynamics. The OU process closely tracks the stochastic decay behavior of SGD gradient norms, supporting continuous-time diffusion approximations.

findings support the interpretation of SGD as a structured stochastic diffusion process governed by anisotropic geometry.

The proposed framework contributes toward a unified mathematical perspective connecting optimization theory, stochastic processes, partial differential equations, and deep learning dynamics.

References

[1] S. Mandt, M. D. Hoffman, and D. M. Blei, "Stochastic Gradient Descent as Approximate Bayesian Inference," *Journal of Machine Learning Research*, vol. 18, no. 134, pp. 1–35, 2017.

[2] N. S. Keskar et al., "On Large-Batch Training for Deep Learning: Generalization Gap and Sharp Minima," *International Conference on Learning Representations*, 2017.

[3] N. Yang, C. Tang, and Y. Tu, "Stochastic Gradient Descent Introduces an Effective Landscape-Dependent Regularization Favoring Flat Solutions," *Physical Review Letters*, vol. 130, no. 23, 2023.

[4] N. Yang et al., "Transient Learning Dynamics Drive Escape from Sharp Valleys in Stochastic Gradient Descent," *arXiv preprint arXiv:2601.10962*, 2026.

[5] D. Granziol and K. Juarev, "Hessian Spectral Analysis at Foundation Model Scale," *arXiv preprint arXiv:2602.00816*, 2026.

[6] Y. Zhang, N. Yang, and Y. Tu, "On the Superlinear Relationship Between SGD Noise Covariance and Loss Landscape Curvature," *arXiv preprint arXiv:2602.05600*, 2026.

[7] B. Øksendal, *Stochastic Differential Equations: An Introduction with Applications*. Springer, 2003.

[8] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.

[9] S. Amari, "Natural Gradient Works Efficiently in Learning," *Neural Computation*, vol. 10, no. 2, pp. 251–276, 1998.

[10] L. Bottou, "Large-Scale Machine Learning with Stochastic Gradient Descent," *Proceedings of COMPSTAT*, pp. 177–186, 2010.